



Figure 6: Usage of IoT

Visualisation is one of the important aspects of the IoT. It helps users to interact with the environment. The continuous development of touchscreen technologies has led to higher usage of smartphones and tablets. This has helped the layman to understand and visualise the technology. With the help of visualisation techniques, more useful and meaningful information can be extracted/analysed and depicted. This improves decision making capabilities.

### Using IoT technology

It can be useful in various day-to-day scenarios as shown in Figure 6.

### Challenges

Like every other technology, there are some challenges with IoT also, such as small organisations finding it difficult to adopt due to budget issues, security risks and so on. Some of the challenges encountered with IoT are:

**Security:** The IoT connects many devices together. Many devices in a network may be less expensive, and might not have been tested for malware and bugs. So in such a situation, a security assurance cannot be given. Moreover, there are many layers of networks, so communication between various devices at the same time can increase the risk factor with this technology.

**Complexity, confusion and integration issues:** As the IoT connects various networks which are based on different protocols, testing and integrating them will be very difficult. Currently, there are no standards available, which creates confusion during integration.

**Privacy:** As there are no universally-accepted compliance parameters available for medical and lifestyle applications, concerns about privacy and trust increase. Multiple users will be accessing the same network, in which case, data security, data accuracy and privacy may get compromised. 

### References

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